Normalization

ReadMore Community Library Catalogue Search Output:

UNF:

CATALOGUE
(catal_call_no,catal_title,catal_contents,catal_pub_id,catal_pubname,catal_date_pub,catal_edition,catal_pageno,catal_lang,catal_note,(author_id,author_name,(sub_id,sub_desc,(isbn,is_type))))

1NF:

CATALOGUE(catal_call_no,catal_title,catal_contents,catal_pub_id,catal_pubname,catal_date_pub,catal_edition,catal_pageno,catal_lang,catal_note)

AUTHOR(catal_call_no ,author_id,author_name)

SUBJECT(catal_call_no ,sub_id,sub_desc)

IDENTIFIER(isbn, is_type)

Candidate keys:

- ➔ catal_call_no ,author_id
- ➔ catal_call_no ,sub_id

Partial dependencies:

author_id -> author_name

sub_id -> sub_desc

2NF:

CATALOGUE(catal_call_no,catal_title,catal_contents,catal_pub_id,catal_pubname,catal_date_pub,catal_edition,catal_pageno,catal_lang,catal_note)

AUTHOR(catal_call_no, author_id)

AUTHOR_DETAILS(author_id,author_name)

SUBJECT_DETAILS(catal_call_no ,sub_id)

SUBJECT(sub_id,sub_desc)

IDENTIFIER(isbn, is_type)

Transitive Dependencies:

catal_pub_id -> catal_pubname

3NF:

CATALOGUE(catal_call_no,catal_title,catal_contents,catal_pub_id,catal_date_pub,catal_edition,catal_pageno,catal_lang,catal_note)

PUBLISHER(catal_pub_id, catal_pubname)

AUTHOR(catal_call_no, author_id)

AUTHOR_DETAILS(author_id,author_name)

SUBJECT_DETAILS(catal_call_no ,sub_id)

SUBJECT(sub_id,sub_desc)

IDENTIFIER(isbn, is_type)

Full Dependencies:

catal_call_no -> catal_title,catal_contents,catal_pub_id,catal_pubname,catal_date_pub,catal_edition,catal_pageno,catal_lang,catal_note

catal_pub_id -> catal_pubname

catal_call_no, author_id ->

catal_call_no ,sub_id ->

author_id -> author_name

sub_id -> sub_desc

isbn -> is_type

ReadMore Community Library Borrowers Quarterly Report:

UNF:

BORROWER(borrowr_no, borrowr _fname, borrowr _lname, borrowr _class, borrowr _stat_pend, borrowr _stat_ban,(bran_code,copy_id,bran_name, borrowed_dat_tym,due_bak,returned_dat_tym,fine_amt, dat_fine_paid))

1NF:

BORROWER(borrowr_no, borrowr _fname, borrowr _lname, borrowr _class, borrowr _stat_pend, borrowr _stat_ban)

LOAN(borrowr_no,bran_code,copy_id,bran_name,borrowed_dat_tym,due_bak,returned_dat_tym,fine_amt, dat_fine_paid)

Candidate Keys:

- ➔ bran_code ,borrowed_dat_tym, borrowr_no
- ➔ bran_code ,borrowed_dat_tym, copy_id

Partial Dependency:

bran_code -> bran_name

2NF:

BORROWER(borrowr_no, borrowr_fname, borrowr_lname, borrowr_class, borrowr_stat_pend, borrowr_stat_ban)

LOAN(borrowr_no,bran_code,copy_id,borrowed_dat_tym,due_bak,returned_dat_tym,fine_amt, dat_fine_paid)

BRANCH(bran_code,bran_name)

Transitive Dependency:

No Transitive Dependency here

3NF:

BORROWER(borrowr_no, borrowr_fname, borrowr_lname, borrowr _class, borrowr_stat_pend, borrowr_stat_ban)

LOAN(borrowr_no,bran_code,copy_id,borrowed_dat_tym,due_bak,returned_dat_tym,fine_amt, dat_fine_paid)

BRANCH(bran_code,bran_name)

Full Dependencies:

borrowr_no -> borrowr_fname,borrowr_lname,borrowr_class,borrowr_stat_pend,borrowr_stat_ban

borrowr_no,bran_code,copy_id,borrowed_dat_tym -> due_bak,returned_dat_tym,fine_amt, dat_fine_paid

bran_code -> bran_name

Combined normalizations:

CATALOGUE(catal_call_no,catal_title,catal_contents,catal_pub_id,catal_date_pub,catal_edition,catal_pageno,catal_lang,catal_note)

PUBLISHER(catal_pub_id, catal_pubname)

AUTHOR(catal_call_no, author_id)

AUTHOR_DETAILS(author_id,author_name)

SUBJECT(sub_id,sub_desc)

SUBJECT_DETAILS(catal_call_no ,sub_id)

IDENTIFIER(isbn, is_type)

BORROWER(borrowr_no, borrowr_fname, borrowr_lname, borrowr _class, borrowr_stat_permit, borrowr_stat_ban)

LOAN(borrowr_no,bran_code,copy_id,borrowed_dat_tym,due_bak,returned_dat_tym,fine_amt, dat_fine_paid)

BRANCH(bran_code,bran_name)

Attribute Synthesis:

CATALOGUE(catal_call_no,catal_title,catal_contents,catal_pub_id,catal_date_pub,catal_edition,catal_pageno,catal_lang,catal_note)

PUBLISHER(catal_pub_id, catal_pubname)

AUTHOR(catal_call_no, author_id, author_name)

SUBJECT(catal_call_no,sub_id,sub_desc)

IDENTIFIER(isbn, is_type)

BORROWER(borrowr_no, borrowr_fname, borrowr_lname, borrowr _class, borrowr_stat_permit, borrowr_stat_ban)

LOAN(borrowr_no,bran_code,copy_id,borrowed_dat_tym,due_bak,returned_dat_tym,fine_amt, dat_fine_paid)

BRANCH(bran_code,bran_name)